Mankei Tsang

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RESEARCH INTERESTS

◇ Quantum Metrology, Quantum Optics, Sensing and Imaging, Superresolution.

SELECTED INFLUENTIAL WORKS

◇ Pioneered the field of *quantum-inspired superresolution* for incoherent imaging.

- First paper of the field: Tsang, Nair, and Lu, Physical Review X **6**, 031033 (2016).
- Review papers: Tsang, Contemporary Physics, **60**, 279–298 (2019); Lvovsky *et al.*, Optica **13**, 2379 (2026).
- See https://blog.nus.edu.sg/mankei/superresolution/ for subsequent work in this field, including 70+ experiments and 2 on-sky demonstrations [Kim *et al.*, ApJL **993**, L3 (2025); Wallis *et al.*, arXiv:2606.18025.]

◇ Proposed the use of *spectral photon counting* to enhance noise spectroscopy with optical interferometers.

- Ng *et al.*, Physical Review A **93**, 042121 (2016).
- Caltech, JPL, and Fermilab scientists will build an experiment based on this idea to detect signatures of quantum gravity under the Gravity from the Quantum Entanglement of Space Time (GQuEST) project; see Sec. III C in Vermeulen *et al.*, Phys. Rev. X **15**, 011034 (2025).

◇ Proposed the use of *cavity quantum electro-optics* for quantum transduction between optical and microwave photons.

- Tsang, Physical Review A **81**, 063837 (2010); **84**, 043845 (2011).
- See https://blog.nus.edu.sg/mankei/cavity-quantum-electro-optics/ for 10+ experiments based on this idea.

RESEARCH AND TEACHING EXPERIENCE

◇ **Associate Professor** (**with tenure**, Jul 2017–present)
Assistant Professor (Aug 2011–Jun 2017)
Department of Electrical and Computer Engineering (joint appointment with Department of Physics), National University of Singapore, Singapore

- Taught courses:
  - PC4246 Quantum Optics (Lectures, 2025–present)
  - PC5286/PC5287 M.Sc. Physics Coursework Project (2024–present)
  - EE6440 Advanced Topics in Photonics (Part II lectures, 2018–present)
  - EE5517 Optical Engineering (Lectures, 2012–present)
  - EE4002/EE4002D/EE4002R B.Eng. Dissertation/Design Capstone/Research Capstone (2012–present)
  - PC2134/PC2174A Mathematical Methods in Physics I (Lectures and tutorials, 2021–2023)
  - EE2012/EE2012A Analytic Methods in ECE (Tutorials, 2019–2022)
  - MA2501 Differential Equations and Systems (Part II lectures and tutorials, 2017–2018)
  - PC2232 Physics for Electrical Engineers (Lectures, 2015–2017)
- Faculty of Engineering Teaching Commendation List for Academic Year 2017/2018.
- Supervised PhD students:
  - Xiaojie Tan (Ph.D. in EE, Aug 2021–Sep 2023)
  - Shilin Ng (Ph.D. in Physics, Jan 2013–May 2017)

    - Shan Zheng Ang (Ph.D. in EE, Aug 2012–Dec 2016)
  - Supervised Master students: Xinyi Sui (M.Sc. in Physics, Aug 2025–May 2026), Wenjie He (M.Sc. in Physics, Aug 2024–May 2025), Kenneth Y. W. Ng (M.Eng. in EE, Jan 2019–Jan 2022), Soham Sataparno Saha (M.Eng. in EE, co-supervised with Aaron Danner, Jan 2014–Aug 2015), Dan Li (M.Sc. in Physics, Aug 2013–Aug 2015).
  - Supervised postdoctoral fellows: Xiaojie Tan (2023–2024), Aparna Bisht (co-supervised with Aaron Danner, 2021–2022), Luo Qi (co-supervised with Aaron Danner, 2019–2024), Wei Wang (co-supervised with Pakorn Kanchanawong, 2020–2024), Lianwei Chen (co-supervised with Minghui Hong and Aaron Danner, 2019–2021), Samuel Barnett (co-supervised with Pakorn Kanchanawong, 2019), Shibdas Roy (2015–2016), Jun Deng (co-supervised with Aaron Danner, 2014–2015), Alexander Crosse (2014–2015), Cong Son Ho (co-supervised with Mansoor B. A. Jalil, 2014–2016), Xiao-Ming Lu (2014–2016), Adam Zaman Chaudhry (2013–2014), Davide Girolami (2013–2014), Sumei Huang (2013–2015), Brent Yen (2012), Andy Chia (2012), and Ranjith Nair (2012–2018).

◇ **Adjunct Research Professor** (Oct 2010–Aug 2011)
Postdoctoral Fellow (Jan 2010–Aug 2011)
Center for Quantum Information and Control (CQuIC), Department of Physics and Astronomy, University of New Mexico, USA

- Advisor: Prof. **Carlton M. Caves**
- Investigated fundamental problems concerning estimation, control, and limits in quantum sensing.

◇ **Postdoctoral Fellow** (Jan 2008–Dec 2009)
Keck Foundation Center for Extreme Quantum Information Theory (xQIT), Research Laboratory of Electronics, Massachusetts Institute of Technology, USA

- Advisors: Prof. **Jeffrey H. Shapiro** and Prof. **Seth Lloyd**
- Developed estimation techniques for quantum sensing and imaging.

◇ **Postdoctoral Scholar** (Jul 2006–Dec 2007)
Graduate Researcher (Sep 2002–Jun 2006)
Optical Information Processing Group, Department of Electrical Engineering, California Institute of Technology, USA

- Advisor: Prof. **Demetri Psaltis**
- performed fundamental and applied research in quantum optics, nano-optics, and nonlinear optics.
- gave lectures for the *Electromagnetic Engineering* course.

◇ **Teaching Assistant** (Sep 2002–Jun 2003)
Department of Electrical Engineering, California Institute of Technology, USA

- gave review lectures, answered students' questions, and graded homeworks for the courses *Fourier Optics*, *Electromagnetic Engineering*, and *Optics in Nano-structures*.

◇ **Undergraduate Researcher** (Jun 2001–Jun 2002)
Electro-Physical Integration Group, Electrical Engineering Department, University of California - Los Angeles, USA

- Advisor: Prof. Elliott R. Brown
- performed analytic modeling and numerical analysis for the Acoustic Dental Imaging Project.

EDUCATION

◇ **Doctor of Philosophy (Ph.D.)** in Electrical Engineering, 2006
Master of Science (M.S.) in Electrical Engineering, 2004
California Institute of Technology, USA (Sep 2002–Jun 2006)

- Advisor: Prof. Demetri Psaltis
- Ph.D. Thesis Title: Classical and Quantum Nonlinear Optical Information Processing

◇ **Bachelor of Science (B.S.)** in Electrical Engineering, 2002
Bachelor of Science (B.S.) in Physics, 2002
University of California–Los Angeles, USA (Oct 1998–Jun 2002)

- Double major, GPA 3.959/4.000, Summa Cum Laude

GRANTS

◇ Quantum Engineering Programme, Singapore National Research Foundation (Apr 2019–Mar 2024)

- Project title: "Quantum Photonics for Superresolution Confocal Fluorescence Microscopy"
- Total direct cost: SGD 2,137,700[1]
- Lead PI (co-PIs: Aaron Danner, Minghui Hong, Pakorn Kanchanawong, Alexander Ling)

◇ Academic Research Fund Tier 1 Grant, Singapore Ministry of Education (Mar 2016–Feb 2019)

- Project title: "Optimal Quantum Photonics for Imaging and Spectroscopy"
- Total direct cost: SGD 170,000
- Sole PI

◇ Singapore National Research Foundation Fellowship (Aug 2011–Aug 2016)

- Project title: "General Theory of Quantum Sensors: Estimation, Control, and Fundamental Limit"
- Total direct cost: SGD 2,818,000
- Sole PI

SERVICES

◇ **Manuscript Referee** for many journals, including Physical Review Letters, Physical Review X, other Physical Review journals, Science Advances, Quantum, New Journal of Physics, The Astrophysical Journal, The Astronomical Journal, Optica, Optics Letters, Optics Express, JOSA A, JOSA B, IEEE Transactions on Automatic Control, IEEE Journal of Selected Topics in Signal Processing, EPJ Quantum Technology, Nature Photonics, Nature Physics, and Nature Communications.

◇ **Proposal Referee** for the US National Science Foundation, the European Research Council, the UK Royal Society, the UK Engineering and Physical Sciences Research Council (EPSRC), the German Research Foundation, the Austrian Science Fund, the Israel Science Foundation, the Hong Kong Research Grants Council, and the Singapore Ministry of Education.

◇ **Advisory Editorial Board** member for Quantum Measurements and Quantum Metrology, De Gruyter (Apr 2012–Dec 2020).

◇ **Organizer and Presider** of the Workshop on Quantum Metrology and Quantum-Inspired Superresolution (virtual).

3. 3rd edition on Aug 4–7, 2026, 24 speakers, ~ 80 participants.
2. 2nd edition on Aug 9–11, 2023, 14 speakers, ~ 60 participants.
1. 1st edition on Jul 1–2, 2021, 16 speakers, ~ 60 participants.

◇ **Program Committee** member for Optica Quantum 2.0 Conference (Boston, Massachusetts, USA, Jun 2022), PRACQSYS 2019 (Hong Kong, China, Dec 2019), and SPIE/COS Photonics Asia (Beijing, China, Oct 2016, Oct 2014).

◇ **Presider** for Optica Quantum 2.0 Conference (Denver, Colorado, USA, Jun 2023), QCMC (Baton Rouge, Louisiana, USA, Mar 2018), PQE (Snowbird, Utah, USA, Jan 2017), PRACQSYS (Sydney, Australia, Jul 2015; Tokyo, Japan, Sep 2012), SPIE/COS Photonics Asia (Beijing, China, Oct 2014), APS March Meeting (Denver, Colorado, USA, Mar 2014), Croucher Advanced Study Institute (Hong Kong, China, Dec 2011), and CLEO/IQEC (Baltimore, Maryland, USA, Jun 2009).

◇ **Graduate Admissions Committee** (2016–2023), **Academic Committee** (2018–2019), and **Social Committee** (2012–present) member in the ECE department of NUS.

◇ **Thesis Committee** member and **Chair of PhD Oral Defence** for many students.

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[1] 1 SGD $\approx$ 1.0 CAD $\approx$ 0.73 USD $\approx$ 0.68 EUR $\approx$ 0.59 GBP as of Nov 2023.

JOURNAL PUBLICATIONS

75. Xinyi Sui and Mankei Tsang,
"Spectrum analysis with quantum dynamical systems. II. Finite-time analysis,"
Physical Review A **114**, 053716 (2026).

74. A. I. Lvovsky, Michael R. Grace, Saikat Guha, Mankei Tsang, Gerardo Adesso, and Nicolas Treps,
"Passive optical superresolution at the quantum limit,"
Optica **13**, 2379–2400 (2026).

    ◇ Review paper.

73. Mankei Tsang,
"Approaching the Ultimate Limit of Quantum Multiparameter Estimation by Many-Body Physics,"
Physical Review Letters **137**, 070804 (2026).

72. Mankei Tsang,
"Quantum reversal: a general theory of coherent quantum absorbers,"
Quantum **9**, 1650 (2025).

71. Mankei Tsang,
"Quantum Onsager relations,"
Quantum Science and Technology **10**, 015015 (2024).

    ◇ Included in Focus Issue on Thermodynamics in Quantum Coherent Platforms.

70. Xiao-Jie Tan and Mankei Tsang,
"Quantum limit to subdiffraction incoherent optical imaging. III. Numerical analysis,"
Physical Review A **108**, 052416 (2023).

69. Mankei Tsang,
"Operational meanings of a generalized conditional expectation in quantum metrology,"
Quantum **7**, 1162 (2023).

68. Xiao-Jie Tan, Luo Qi, Lianwei Chen, Aaron J. Danner, Pakorn Kanchanawong, and Mankei Tsang,
"Quantum-inspired superresolution for incoherent imaging,"
Optica **10**, 1189–1194 (2023).

67. Mankei Tsang,
"Efficient superoscillation measurement for incoherent optical imaging,"
IEEE Journal of Selected Topics in Signal Processing **11**, 513–524 (2023).

    ◇ Included in Special Issue on Recent Advances in Wideband Signal Processing for Classical and Quantum Synthetic Apertures.

66. Mankei Tsang,
"Quantum noise spectroscopy as an incoherent imaging problem,"
Physical Review A **107**, 012611 (2023).

65. Mankei Tsang,
"Generalized conditional expectations for quantum retrodiction and smoothing,"
Physical Review A **105**, 042213 (2022).

64. Mankei Tsang,
"Quantum limit to subdiffraction incoherent optical imaging. II. A parametric-submodel approach,"
Physical Review A **104**, 052411 (2021).

63. Mankei Tsang,
"Poisson Quantum Information,"
Quantum **5**, 527 (2021).

    ◇ Selected for a Perspective article on the *Quantum Views* website [1].

62. Mankei Tsang,
"Physics-inspired forms of the Bayesian Cramér-Rao bound,"
Physical Review A **102**, 062217 (2020).

◇ Editors' Suggestion [2].

61. Mankei Tsang, Francesco Albarelli, and Animesh Datta,
"Quantum Semiparametric Estimation,"
Physical Review X **10**, 031023 (2020).

60. Mankei Tsang,
"Resolving starlight: a quantum perspective,"
Contemporary Physics **60**, 279–298 (2019).

◇ Invited review paper.

59. Mankei Tsang,
"Semiparametric estimation for incoherent optical imaging,"
Physical Review Research **1**, 033006 (2019).

58. Mankei Tsang and Ranjith Nair,
"Resurgence of Rayleigh's curse in the presence of partial coherence: comment,"
Optica **6**, 400 (2019).

57. Mankei Tsang,
"Quantum limit to subdiffraction incoherent optical imaging,"
Physical Review A **99**, 012305 (2019).

◇ Editors' Suggestion [2].

◇ Covered by the APS *Physics Buzz* website [3].

56. Mankei Tsang,
"Subdiffraction incoherent optical imaging via spatial-mode demultiplexing: Semiclassical treatment,"
Physical Review A **97**, 023830 (2018).

55. Mankei Tsang,
"Quantum limits on the time-bandwidth product of an optical resonator,"
Optics Letters **43**, 150 (2018).

54. Fan Yang, Ranjith Nair, Mankei Tsang, Christoph Simon, and Alexander I. Lvovsky,
"Fisher information for far-field linear optical superresolution via homodyne or heterodyne detection in a higher-order local oscillator mode,"
Physical Review A **96**, 063829 (2017).

53. Mankei Tsang,
"Conservative classical and quantum resolution limits for incoherent imaging,"
Journal of Modern Optics **65**, 1385–1391 (2018).

◇ Special issue for the 47th Winter Colloquium on the Physics of Quantum Electronics (PQE).

52. Shan Zheng Ang, Ranjith Nair, and Mankei Tsang,
"Quantum limit for two-dimensional resolution of two incoherent optical point sources,"
Physical Review A **95**, 063847 (2017).

51. Mankei Tsang,
"Subdiffraction incoherent optical imaging via spatial-mode demultiplexing,"
New Journal of Physics **19**, 023054 (2017).

50. Ranjith Nair and Mankei Tsang,
"Far-field Superresolution of Thermal Electromagnetic Sources at the Quantum Limit,"
Physical Review Letters **117**, 190801 (2016).

◇ Editors' Suggestion [4].

49. Mankei Tsang, Ranjith Nair, and Xiao-Ming Lu,
"Quantum Theory of Superresolution for Two Incoherent Optical Point Sources,"
Physical Review X **6**, 031033 (2016).

◇ Selected for a Viewpoint article on the APS *Physics* website [5, 6].

◇ Covered by the APS *Physics Buzz* website [7], the IoP *Physics World* website [8], and the *Phys.org* website [9, 10].

- ◇ Mentioned by articles on the OSA *Optics & Photonics News* website [11], the AAAS *EurekAlert!* website [12], and the *N+1* website [13, 14].
- ◇ Chosen as #2 of top physics breakthroughs of 2016 by the Foundational Questions Institute (FQXi) podcast [15].

48. Xiao-Ming Lu and Mankei Tsang,
"Quantum Weiss-Weinstein bounds for quantum metrology,"
Quantum Science and Technology **1**, 015002 (2016).

47. Shilin Ng, Shan Zheng Ang, Trevor A. Wheatley, Hidehiro Yonezawa, Akira Furusawa, Elanor H. Huntington, and Mankei Tsang,
"Spectrum analysis with quantum dynamical systems,"
Physical Review A **93**, 042121 (2016).

46. Ranjith Nair and Mankei Tsang,
"Interferometric superlocalization of two incoherent optical point sources,"
Optics Express, **24**, 3684 (2016).

    - ◇ Covered by a news article in the *Laser Focus World* magazine [16].

45. Shawn Y. Siew, Soham S. Saha, Mankei Tsang, and Aaron J. Danner,
"Rib Microring Resonators in Lithium Niobate on Insulator,"
IEEE Photonics Technology Letters **28**, 573 (2016).

44. Mankei Tsang,
"Volterra filters for quantum estimation and detection,"
Physical Review A **92**, 062119 (2015).

43. Dominic W. Berry, Mankei Tsang, Michael J. W. Hall, and Howard M. Wiseman,
"The quantum Bell-Ziv-Zakai bounds and Heisenberg limits for waveform estimation,"
Physical Review X **5**, 031018 (2015).

42. Ranjith Nair and Mankei Tsang,
"Quantum Optimality of Photon Counting for Temperature Measurement of Thermal Astronomical Sources,"
The Astrophysical Journal **808**, 125 (2015).

41. Mankei Tsang,
"Quantum limits to optical point-source localization,"
Optica **2**, 646 (2015).

40. Trevor A. Wheatley, Mankei Tsang, Ian R. Petersen, and Elanor H. Huntington,
"Improved mirror position estimation using resonant quantum smoothing,"
EPJ Quantum Technology **2**, 13 (2015).

39. Shilin Ng and Mankei Tsang,
"Optimal signal processing for continuous qubit readout,"
Physical Review A **90**, 022325 (2014).

38. Mankei Tsang,
"Microwave photonics: Optomechanics sets the beat,"
Nature Physics **10**, 245–246 (2014).

    - ◇ Invited News & Views article.

37. Mankei Tsang,
"Testing quantum mechanics: a statistical approach,"
Quantum Measurements and Quantum Metrology **1**, 84–109 (2013).

    - ◇ Invited.

36. Shan Zheng Ang, Glen I. Harris, Warwick P. Bowen, and Mankei Tsang,
"Optomechanical parameter estimation,"
New Journal of Physics **15**, 103028 (2013).

35. Kohjiro Iwasawa, Kenzo Makino, Hidehiro Yonezawa, Mankei Tsang, Aleksandar Davidovic, Elanor Huntington, and Akira Furusawa,
"Quantum-Limited Mirror-Motion Estimation,"
Physical Review Letters **111**, 163602 (2013).

34. Mankei Tsang,
"Quantum transition-edge detectors,"
Physical Review A **88**, 021801(R) (2013).

    ◇ Rapid Communication.

33. Mankei Tsang,
"Quantum metrology with open dynamical systems,"
New Journal of Physics **15**, 073005 (2013).

32. Mankei Tsang and Ranjith Nair,
"Fundamental quantum limits to waveform detection,"
Physical Review A **86**, 042115 (2012).

31. Mankei Tsang and Carlton M. Caves,
"Evading Quantum Mechanics: Engineering a Classical Subsystem within a Quantum Environment,"
Physical Review X **2**, 031016 (2012).

    ◇ Covered by an article on the *Ars Technica* website [17].

30. Mankei Tsang,
"Ziv-Zakai Error Bounds for Quantum Parameter Estimation,"
Physical Review Letters **108**, 230401 (2012).

29. Mankei Tsang,
"Continuous Quantum Hypothesis Testing,"
Physical Review Letters **108**, 170502 (2012).

    ◇ Editors' Suggestion [4].

    ◇ Selected for a Synopsis on the APS *Physics* website [6, 18].

28. Mankei Tsang,
"Quantum Nonlocality in Weak-Thermal-Light Interferometry,"
Physical Review Letters **107**, 270402 (2011).

27. Mankei Tsang,
"Cavity quantum electro-optics. II. Input-output relations between traveling optical and microwave fields,"
Physical Review A **84**, 043845 (2011).

26. Mankei Tsang, Howard M. Wiseman, and Carlton M. Caves,
"Fundamental Quantum Limit to Waveform Estimation,"
Physical Review Letters **106**, 090401 (2011).

25. Laura Waller, Mankei Tsang, Sameera Ponda, Se Young Yang, and George Barbastathis,
"Phase and amplitude imaging from noisy images by Kalman filtering,"
Optics Express **19**, 2805 (2011).

24. Mankei Tsang and Carlton M. Caves,
"Coherent Quantum-Noise Cancellation for Optomechanical Sensors,"
Physical Review Letters **105**, 123601 (2010).

    ◇ Featured in the "Editors' Choice" section of the *Science* magazine [19].

23. Mankei Tsang,
"Cavity quantum electro-optics,"
Physical Review A **81**, 063837 (2010).

22. Mankei Tsang,
"Optimal waveform estimation for classical and quantum systems via time-symmetric smoothing. II. Applications to atomic magnetometry and Hardy's paradox,"
Physical Review A **81**, 013824 (2010).

21. Mankei Tsang,
"Optimal waveform estimation for classical and quantum systems via time-symmetric smoothing,"
Physical Review A **80**, 033840 (2009).

20. Mankei Tsang,
"Time-Symmetric Quantum Theory of Smoothing,"
Physical Review Letters **102**, 250403 (2009).

19. Mankei Tsang,
"Quantum Imaging beyond the Diffraction Limit by Optical Centroid Measurements,"
Physical Review Letters **102**, 253601 (2009).

    ◇ Editors' Suggestion [4].

    ◇ Selected for a Viewpoint article on the APS *Physics* website [6, 20].

18. Mankei Tsang, Jeffrey H. Shapiro, and Seth Lloyd,
"Quantum theory of optical temporal phase and instantaneous frequency. II. Continuous time limit and state-variable approach to phase-locked loop design,"
Physical Review A **79**, 053843 (2009).

17. Mankei Tsang, Jeffrey H. Shapiro, and Seth Lloyd,
"Quantum theory of optical temporal phase and instantaneous frequency,"
Physical Review A **78**, 053820 (2008).

16. Mankei Tsang,
"Fundamental Quantum Limit to Multiphoton Absorption Rate for Monochromatic Light,"
Physical Review Letters **101**, 033602 (2008).

15. Mankei Tsang and Demetri Psaltis,
"Magnifying perfect lens and superlens design by coordinate transformation,"
Physical Review B **77**, 035122 (2008).

14. Ye Pu, Jie Wu, Mankei Tsang, and Demetri Psaltis,
"Optical parametric generation in periodically poled $KTiOPO_4$ via extended phase matching,"
Applied Physics Letters **91**, 131120 (2007).

13. Mankei Tsang and Demetri Psaltis,
"Theory of resonantly enhanced near-field imaging,"
Optics Express **15**, 11959 (2007).

12. Mankei Tsang,
"Decoherence of quantum-enhanced timing accuracy,"
Physical Review A **75**, 063809 (2007).

11. Mankei Tsang,
"Relationship between resolution enhancement and multiphoton absorption rate in quantum lithography,"
Physical Review A **75**, 043813 (2007).

10. Mankei Tsang and Demetri Psaltis,
"Reflectionless evanescent wave amplification via two dielectric planar waveguides,"
Optics Letters **31**, 2741 (2006); Erratum: **32**, 86 (2007).

9. Mankei Tsang,
"Quantum Temporal Correlations and Entanglement via Adiabatic Control of Vector Solitons,"
Physical Review Letters **97**, 023902 (2006); Erratum: **99**, 049901(E) (2007).

8. Mankei Tsang,
"Spectral phase conjugation via extended phase matching,"
Journal of the Optical Society of America B **23**, 861 (2006).

7. Mankei Tsang and Demetri Psaltis,
"Propagation of temporal entanglement,"
Physical Review A **73**, 013822 (2006).

6. Martin Centurion, Ye Pu, Mankei Tsang, and Demetri Psaltis,
"Dynamics of filament formation in a Kerr medium,"
Physical Review A **71**, 063811 (2005); Erratum: **74**, 069902(E) (2006).

5. Mankei Tsang and Demetri Psaltis,
"Spontaneous spectral phase conjugation for coincident frequency entanglement,"
Physical Review A **71**, 043806 (2005).

4. Mankei Tsang and Demetri Psaltis,
"Spectral phase conjugation by quasi-phase-matched three-wave mixing,"
Optics Communications **242**, 659 (2004).

3. Mankei Tsang and Demetri Psaltis,
"Spectral phase conjugation with cross-phase modulation compensation,"
Optics Express **12**, 2207 (2004).

2. Mankei Tsang, Demetri Psaltis, and Fiorenzo G. Omenetto,
"Reverse propagation of femtosecond pulses in optical fibers,"
Optics Letters **28**, 1873 (2003).

1. Mankei Tsang and Demetri Psaltis,
"Dispersion and nonlinearity compensation by spectral phase conjugation,"
Optics Letters **28**, 1558 (2003).

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[1] Cosmo Lupo, "Poisson states in quantum information," Quantum Views **5**, 59 (2021), https://doi.org/10.22331/qv-2021-09-02-59.
[2] PRA Editors' Suggestions are "a small number of papers published in Physical Review A that the editors and referees find of particular interest, importance, or clarity;" see http://doi.org/10.1103/PhysRevA.88.020001.
[3] Kendra Redmond, "We're Nowhere Near the Limit on Telescope Resolution, According to New Physics," APS Physics Central – Physics Buzz Blog (2018), https://web.archive.org/web/20181213122714/http://physicsbuzz.physicscentral.com/2018/12/were-nowhere-near-limit-on-telescope.html, retrieved Dec 13, 2018.
[4] To promote reading across fields, the editors of Physical Review Letters offer "Suggestions" each week of papers that they hope will lead readers to explore other areas of physics; see http://doi.org/10.1103/PhysRevLett.118.030001.
[5] Gabriel Durkin, "Viewpoint: Unlocking the hidden information in starlight," Physics **9**, 100 (2016), https://doi.org/10.1103/Physics.9.100.
[6] *Physics* is an online American Physical Society publication that "spotlights exceptional research;" see http://physics.aps.org/ and https://doi.org/10.1103/PhysRevLett.101.120001.
[7] Kendra Redmond, "Resolving Starlight with Quantum Technology," APS Physics Central – Physics Buzz Blog (2016), https://web.archive.org/web/20160902185219/http://physicsbuzz.physicscentral.com/2016/08/resolving-starlight-with-quantum.html, retrieved Sep 2, 2016.
[8] Edwin Cartlidge, "Tapping into light's hidden information to push fundamental diffraction limit," IoP Physics World and nanotechweb.org (2016), https://physicsworld.com/a/tapping-into-lights-hidden-information-to-push-fundamental-diffraction-limit/, retrieved Jul 16, 2022.
[9] Bob Yirka, "Quantum mechanics technique allows for pushing past 'Rayleigh's curse'," Phys.org (2016), https://phys.org/news/2016-09-quantum-mechanics-technique-rayleigh-curse.html, retrieved Jul 16, 2022.
[10] Ingrid Fadelli, "A new theory to test hypotheses and methods for exoplanet detection," Phys.org (2021), https://phys.org/news/2021-10-theory-methods-exoplanet.html, retrieved Jul 16, 2022.
[11] Patricia Daukantas, "Beating Rayleigh with quantum mechanics," OSA Optics & Photonics News (2016), https://www.optica-opn.org/news/rayleigh-limit/, retrieved Jul 16, 2022.
[12] University of Toronto, "Physicists harness neglected properties of light," AAAS EurekAlert! (2017), https://www.eurekalert.org/news-releases/573130, retrieved Jul 16, 2022.
[13] Oksana Borzenkova, "Ultra-high resolution in microscopy made possible without fluorescence," N+1 (2021), https://nplus1.ru/news/2021/05/21/Super-resolution-imaging, in Russian, retrieved Jul 16, 2022.
[14] Alexander Dubov, "Invisible—physicist Alexander Lvovsky on how to look beyond the limit of diffraction," N+1 (2021), https://nplus1.ru/material/2021/06/30/lvovsky-enhance, in Russian, retrieved Jul 16, 2022.
[15] The Foundational Questions Institute (FQXi), "December 31, 2016 Podcast – 2016 Countdown Part 2," Online (2016), https://fqxi.org/community/podcast/2016.12.31, retrieved Jul 16, 2022.
[16] John Wallace, "SLIVER to determine separation of two sources closer than Rayleigh limit," Laser Focus World **52** (2016), https://www.laserfocusworld.com/test-measurement/test-measurement/article/16546922/sliver-to-determine-separation-of-two-sources-closer-than-rayleigh-limit, retrieved Jul 16, 2022.
[17] Chris Lee, "Demolishing Heisenberg with clever math and experiments," Ars Technica (2012), http://arstechnica.com/science/2012/09/demolishing-heisenberg-with-clever-math-and-experiments/, retrieved Jul 16, 2022.
[18] David Voss, "Synopsis: Quantum Testing," Physics **5**, s65 (2012), https://physics.aps.org/articles/v5/s65.
[19] Ian S. Osborne, "Calling for a Quantum Hush," Science **330**, 13 (2010), https://doi.org/10.1126/science.2010.330.6000.twil.
[20] Petr M. Anisimov and Jonathan P. Dowling, "Viewpoint: Super resolution with superposition," Physics **2**, 52 (2009), http://doi.org/10.1103/Physics.2.52.

CONFERENCES

Each **bold name** in this section denotes the presenter.

Invited

42. **Mankei Tsang**,
"Quantum estimation with infinitely many parameters,"
Oral Presentation, WE-Heraeus Seminar on Multiparameter Quantum Sensing and Metrology, Bad Honnef, Germany, Feb 4, 2025.

41. **Mankei Tsang**,
"Quantum noise spectroscopy as an incoherent imaging problem,"
Oral Presentation, the 53rd Winter Colloquium on the Physics of Quantum Electronics (PQE), Snowbird, Utah, USA, Jan 10, 2024.

40. **Mankei Tsang**,
"Quantum noise spectroscopy,"
Workshop on Continuously Monitored Quantum Systems (CMONS 2023), Warsaw, Poland, Dec 13, 2023.

39. **Mankei Tsang**,
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Oral Presentation, Kavli Institute for Theoretical Physics (KITP) Conference: Frontiers of Quantum Metrology, Santa Barbara, California, USA, Oct 12, 2023.

38. **Mankei Tsang**,
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36. **Mankei Tsang**,
"Quantum nonlocality in stellar interferometry,"
Oral Presentation, Quantum-Enhanced Telescopy Workshop, Denver, Colorado, USA, Jun 18, 2023.

35. **Mankei Tsang**,
"Resolving starlight: a quantum perspective,"
Oral Presentation (virtual), the 51st Winter Colloquium on the Physics of Quantum Electronics (PQE), Snowbird, Utah, USA, Jan 12, 2022.

34. **Mankei Tsang**, Francesco Albarelli, and Animesh Datta,
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Oral Presentation (virtual), Paris-Singapore-Tokyo Workshop on Quantum Metrology, Networks and Cryptography, Online, May 21, 2021.

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"Resolving starlight: a quantum perspective,"
Oral Presentation, the First Quantum Science, Engineering and Technology Conference (qSET), Canberra, Australia, Apr 9, 2019.

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"Quantum-inspired measurements for subdiffraction incoherent optical imaging,"
Oral Presentation, the Fifth Workshop on Scalable Information Processing with Quantum Nano-Photonics (SIPQNP), Tucson, Arizona, USA, Mar 26, 2019.

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"Quantum metrology for subdiffraction incoherent optical imaging,"
Oral Presentation, Symposium on Novel Perspectives in Quantum Metrology, German Physical Society (DPG) Spring Meeting, Rostock, Germany, Mar 15, 2019, SYQM 1.1.

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"Seize the moments: enhancing moment estimation for subdiffraction incoherent imaging,"
Oral Presentation, the Fourteenth Singapore-China Physics Frontier Symposium, Shanghai, China, Sep 26, 2018.

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"Subdiffraction incoherent imaging: statistical and quantum approaches,"
Oral Presentation, the 39th Progress in Electromagnetics Research Symposium (PIERS), Singapore, Nov 21, 2017.

26. Ranjith Nair, Xiao-Ming Lu, Shan Zheng Ang, and **Mankei Tsang**,
"Quantum information kills Rayleigh's criterion,"
Oral Presentation, Conference on Quantum Information and Quantum Control VII, Toronto, Canada, Aug 29, 2017.

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"Quantum theory of superresolution for incoherent optical imaging,"
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Plenary Talk, the 47th Winter Colloquium on the Physics of Quantum Electronics, Snowbird, Utah, USA, Jan 12, 2017.

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Oral Presentation, Recent Advances in Quantum Metrology Workshop, Warsaw, Poland, Mar 4, 2016.

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"Quantum spectroscopy,"
Oral Presentation, the Tenth International Workshop on Principles and Applications of Control in Quantum Systems (PRACQSYS), Sydney, Australia, Jul 20, 2015.

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15. **Mankei Tsang**,
"Quantum theory of optical sensing: estimation, control, and fundamental limits,"
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"Continuous quantum hypothesis testing,"
Keynote Talk, the Third Biennial China-Australia Quantum Control Workshop, Brisbane, Australia, Sep 30, 2014.

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"Quantum transition-edge detectors,"
Oral Presentation, the Eighth International Workshop on Principles and Applications of Control in Quantum Systems (PRACQSYS), Monterey, California, USA, Aug 23, 2013.

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"Quantum Microwave Photonics,"
Oral Presentations, the Seventh Winter School on Quantum Information Science, National Cheng Kung University, Taiwan, Republic of China, Feb 28–Mar 2, 2013.

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Oral Presentation, the Fourth Workshop on Quantum Information Science, Hong Kong, China, Dec 28, 2012.

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Oral Presentation, IEEE Lasers and Electro-Optics Society (LEOS) Annual Meeting, Sydney, Australia, Oct 1, 2005, paper TuEE2.

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"Resolving starlight: a quantum perspective,"
Oral Presentation, Institute of Physics Singapore Meeting 2021, Singapore, Oct 1, 2021, code T13.75.

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"Beating Rayleigh's criterion: superresolution of thermal sources with linear optics,"
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"Integrated nonlinear optics: lithium niobate-on-insulator waveguides and resonators,"
Oral Presentation, Integrated Optics: Devices, Materials, and Technologies XXI, SPIE Photonics West, San Francisco, California, Jan 28, 2017.

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Poster Presentation, the Thirteenth International Conference on Quantum Communication, Measurement and Computing (QCMC), Singapore, Jul 5, 2016, paper P2-195.

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Poster Presentation, the Thirteenth International Conference on Quantum Communication, Measurement and Computing (QCMC), Singapore, Jul 5, 2016, paper P2-153.

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"Rib microring resonators in lithium niobate on insulator,"
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"Mismatched quantum filtering and entropic information,"
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INVITED COLLOQUIA AND SEMINARS

37. "Quantum-Inspired Incoherent Imaging and Optomechanical Sensing,"
IQIM Seminar, California Institute of Technology, USA, Jan 12, 2024.

36. "Quantum Waveform Estimation, Detection, and Noise Spectroscopy,"
CQIQC/QO/AMO Seminar, Department of Physics, University of Toronto, Canada, Nov 24, 2023.

35. "Quantum Conditional Expectations,"
CQIQC/QO/AMO Seminar, Department of Physics, University of Toronto, Canada, Nov 10, 2023.

34. "Resolving Starlight: A Quantum Perspective,"
PME Distinguished Quantum Colloquium, University of Chicago, USA, Nov 7, 2023.

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CQIQC/QO/AMO Seminar, Department of Physics, University of Toronto, Canada, Oct 20, 2023.

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Seminar, Kavli Institute for Theoretical Physics (KITP) Program: New Directions in Quantum Metrology, University of California–Santa Barbara, USA, Sep 13, 2023.

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30. "Resolving Starlight: A Quantum Perspective,"
Atomic and Laser Physics Seminar, University of Oxford, UK, Dec 2, 2022.

29. "Resolving Starlight: A Quantum Information Perspective,"
Quantum Information and Metrology Seminar (virtual), University of Nottingham, UK, Jul 27, 2022.

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27. "Resolving Starlight: A Quantum Perspective,"
AP 483 Seminar (virtual), Ginzton Laboratory, Stanford University, USA, Nov 16, 2020.

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Special Lecture, College of Optical Sciences, University of Arizona, USA, Mar 28, 2019.

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Seminar, Kastler Brossel Laboratory, Paris, France, Jul 3, 2018.

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Lecture, University of Nottingham, UK, Jun 13, 2016.

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Seminar, Department of EECS, University of California - Berkeley, USA, Jun 2, 2016.

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Seminar, Centre for Quantum Technologies, Singapore, Nov 18, 2015.

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Seminar, Institute of Modern Optics, Peking University, Beijing, China, Oct 12, 2014.

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Seminar, Center for Quantum Information and Control, University of New Mexico, USA, Feb 27, 2014.

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"Quantum Waveform Detection Theory,"
"Quantum Waveform Estimation Theory,"
Seminars, RIKEN Institute, Wako, Japan, Feb 14, 21, 25, 2013.

15. "Continuous Quantum Hypothesis Testing,"
Colloquium, Department of Applied Mathematics, the Hong Kong Polytechnic University, China, Jan 7, 2013.

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Seminar, A*STAR Data Storage Institute, Singapore, Apr 20, 2012.

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Seminar, Centre for Quantum Technologies, Singapore, Oct 18, 2011.

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Seminar, Department of Electrical and Computer Engineering, National University of Singapore, Singapore, Feb 28, 2011.

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Seminar, JILA, Boulder, Colorado, USA, Feb 21, 2011.

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4. "Fundamental Quantum Limit to Waveform Estimation,"
   Seminar, Center for Quantum Dynamics, Griffith University, Brisbane, Australia, Jul 27, 2010.

3. "Reversing Time and Space in Classical and Quantum Optics,"
   Seminar, Center for Quantum Information and Control, University of New Mexico, USA, Oct 8, 2009.

2. "Beating Classical and Quantum Limits in Optics,"
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